11

HART protocol

The highway addressable remote transducer (HART) protocol is one of a number of smart instrumentation protocols designed for collecting data from instruments, sensors, and actuators by digital communication techniques.

Objectives

When you have completed studying this chapter you will be able to:

- Describe the origin and benefits of the HART protocol
- Describe the three OSI layers of the HART protocol

11.1 Introduction to HART and smart instrumentation

Smart (or intelligent) instrumentation protocols are designed for applications where actual data is collected from instruments, sensors, and actuators by digital communication techniques. These components are linked directly to programmable logic controllers (PLCs) and computers.

The HART (highway addressable remote transducer) protocol is a typical smart instrumentation Fieldbus that can operate in a hybrid 4–20 mA digital fashion.

HART is, by no means, the only protocol in this sphere. There are hundreds of smart implementations produced by various manufacturers – for example Honeywell, which compete with HART. This chapter deals specifically with HART. For information about the other Fieldbus protocols refer to Chapter 12.

At a basic level, most smart instruments provide core functions such as:

- Control of range/zero/span adjustments
- Diagnostics to verify functionality
- Memory to store configuration and status information (such as tag numbers etc.)

Accessing these functions allows major gains in the speed and efficiency of the installation and maintenance process. For example, the time consuming 4–20 mA loop

check phase can be achieved in minutes, and the device can be readied for use in the process by zeroing and adjustment for any other controllable aspects such as the damping value.

11.2 Highway addressable remote transducer (HART)

This protocol was originally developed by Rosemount and is regarded as an open standard, available to all manufacturers. Its main advantage is that it enables an instrumentation engineer to keep the existing 4–20 mA instrumentation cabling and to use simultaneously the same wires to carry digital information superimposed on the analog signal. This enables most companies to capitalize on their existing investment in 4–20 mA instrumentation cabling and associated systems; and to add the further capability of HART without incurring major costs.

HART is a hybrid analog and digital protocol, as opposed to most Fieldbus systems, which are purely digital.

The HART protocol uses the frequency shift keying (FSK) technique based on the Bell 202 communications standard. Two individual frequencies of 1200 and 2200 Hz, representing digits 1 and 0 respectively, are used. The average value of the sine wave (at the 1200 and 2200 Hz frequencies), which is superimposed on the 4–20 mA signal, is zero. Hence, the 4–20 mA analog information is not affected.

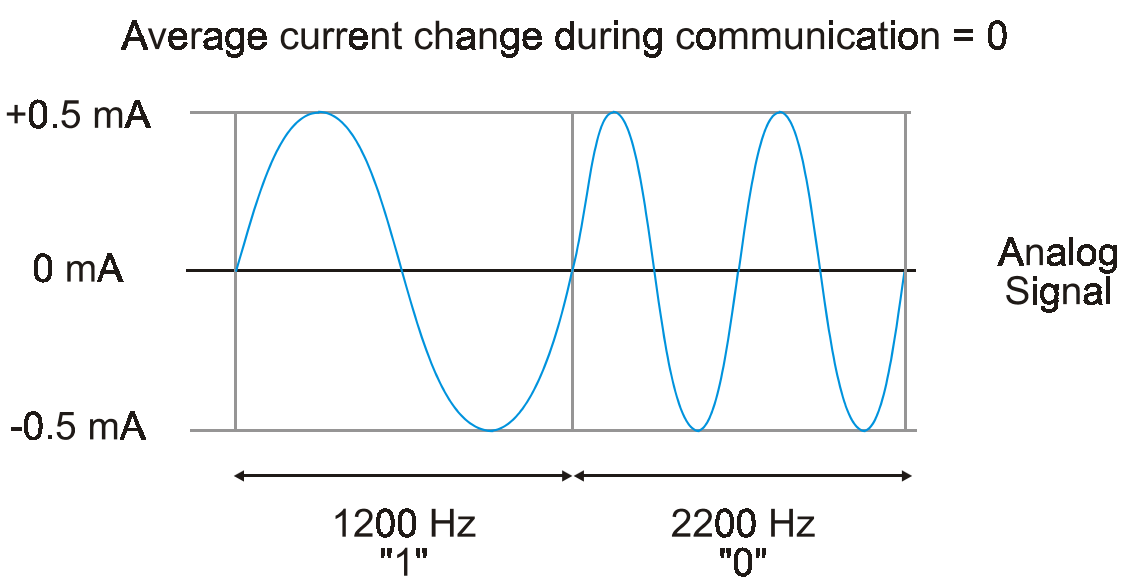


Figure 11.1
Frequency allocation of HART protocol

The HART protocol can be used in three ways:

- In conjunction with the 4–20 mA current signal in point-to-point mode
- In conjunction with other field devices in multidrop mode
- In point-to-point mode with only one field device broadcasting in burst mode

Traditional point-to-point loops use zero for the smart device polling address. Setting the smart device polling address to a number greater than zero creates a multidrop loop.

The smart device then sets its analog output to a constant 4 mA and communicates only digitally.

The HART protocol has two formats for digital transmission of data:

- Poll/response mode
- Burst (or broadcast) mode

In the poll/response mode the master polls each of the smart devices on the highway and requests the relevant information. In burst mode the field device continuously transmits process data without the need for the host to send request messages. Although this mode is fairly fast (up to 3.7 times/second) it cannot be used in multidrop networks.

The protocol is implemented with the OSI model (see Chapter 9**)** using layers 1, 2 and 7. The actual implementation is covered in this chapter.

11.3 Physical layer

The physical layer of the HART Protocol is based on two methods of communication.

- Analog 4–20 mA
- Digital frequency shift keying (FSK)

Analog 4 to 20 mA communications

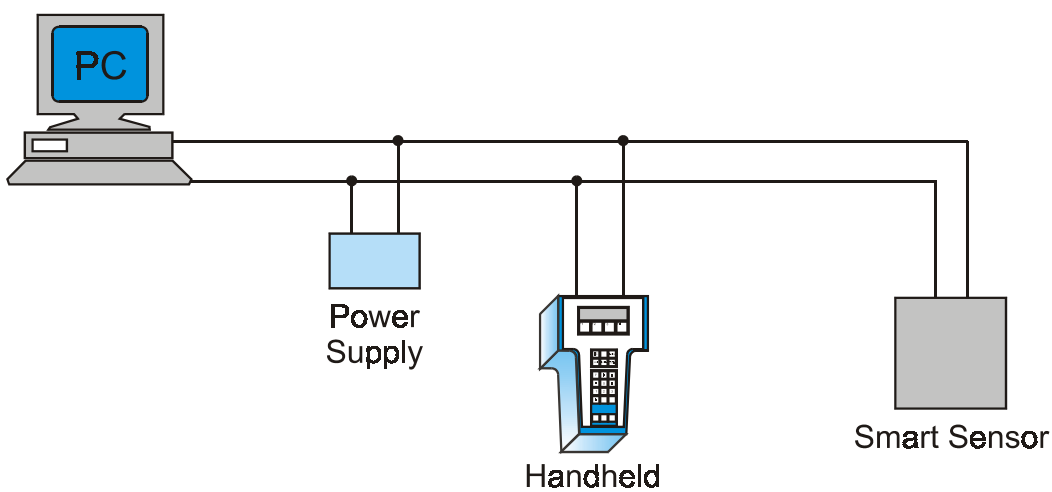


Figure 11.2
HART point-to-point communications

The basic communication of the HART protocol is the 4–20 mA current system. This analog system is used by the sensor to transmit an analog value to the HART PLC or HART card in a PC. In a 4–20 mA the sensor outputs a current value somewhere between 4 and 20 mA that represents the analog value of the sensor. For example, a water tank that is half full – say 3400 kilolitres – would put out 12 mA. The receiver would interpret this 12 mA as 3400 kilolitres. This communication is always point-to-point, i.e. from one device to one other. It is not possible to do multidrop communication using this method alone. If two or more devices put some current on the line at the same time, the resulting current value would not be valid for either device.

Digital multidrop communications

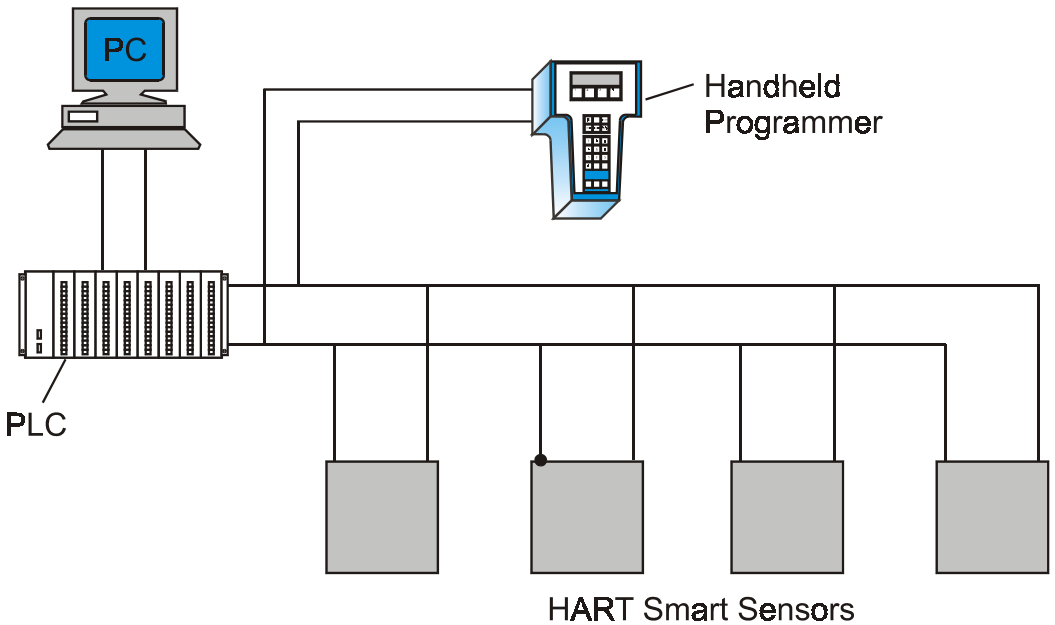


Figure 11.3
HART multi-point communications

For multidrop communications, the HART protocol uses a digital/analog modulation technique known as frequency shift keying (FSK). This technique is based on the Bell 202 communication standard. Data transfer rate is 1200 baud with a digital '0' frequency (2200 Hz) and a digital '1' frequency (1200 Hz). Category 5 shielded, twisted pair wire is recommended by most manufacturers. Devices can be powered by the bus or individually. If the bus powers the devices, only 15 devices can be connected. As the average DC current of an ac frequency is zero, it is possible to place a 1200 Hz or 2200 Hz tone on top of a 4–20 mA signal. The HART protocol does this to allow simultaneous communications on a multidrop system.

The HART handheld communicator

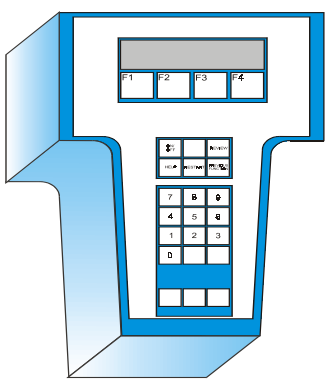


Figure 11.4
HART handheld controller

The HART system includes a handheld control device. This device can be a second master on the system. It is used to read, write, range and calibrate devices on the bus. It can be taken into the field and used for temporary communications. The battery operated handheld has a display and key input for specific commands.

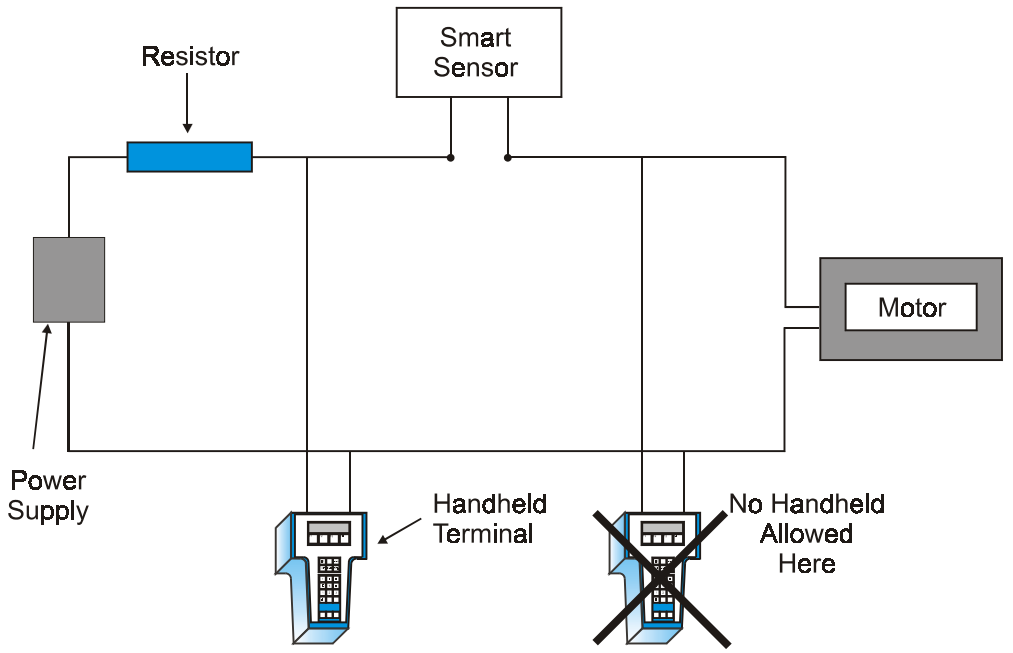


Figure 11.5
HART handheld connection method

The HART field controller in Figure 11.5 is wired in series with the field device (valve positioner or other actuator). In some cases, a bypass capacitor may be required across the terminals of the valve positioner to keep the positioner's series impedance below the 100 Ω level required by HART specifications. Communications with the field controller requires the communicating device (handheld terminal or PC) to be connected across a loop impedance of at least 230 Ω. Communications is not possible across the terminals of the valve positioner because of its low impedance (100 Ω). Instead, the communicating device must be connected across the transmitter or the current sense resistor. (Taken from the HART applications guide by the HART Communications Foundation 1999 *www.hartcomm.org*.)

11.4 Data link layer

The data link frame format is shown in Figure 11.7.

Layer		Description	HART ™
7	Application	Serves up formatted data	Hart Commands
6	Pesentation	Translates Data	
5	Session	Controls Dialogue	
4	Transport	Ensures Message Integrity	
3	Network	Routes Information	
2	Data Link	Detects Errors	Protocol Rules
1	Physical	Connects Device	Bell 202

Figure 11.6
HART protocol implementation of OSI layer model

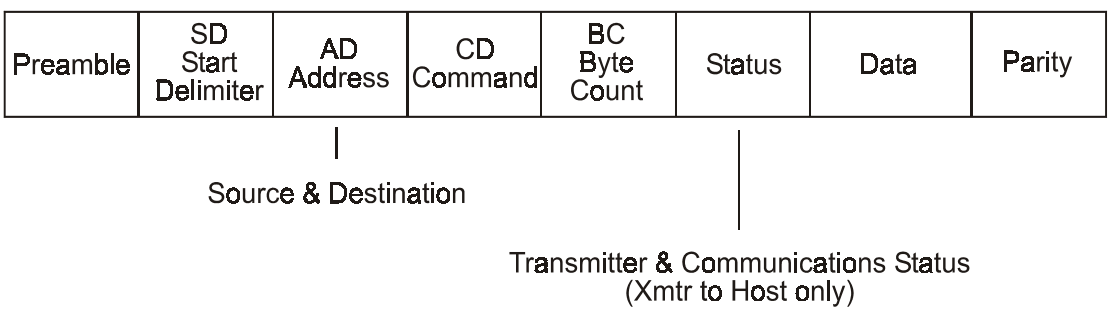


Figure 11.7
HART data link frame format

Two-dimensional error checking, including both vertical and longitudinal parity checks, is implemented in each frame. Each character or frame of information has the following parameters:

- 1 start bit
- 8 data bits
- 1 odd parity bit
- 1 stop bit

11.5 Application layer

The application layer allows the host device to obtain and interpret field device data. There are three classes of commands:

- Universal commands
- Common practice commands
- Device specific commands

Examples of these commands are listed below.

Universal commands

- Read manufacturer and device type
- Read primary variable (PV) and units
- Read current output and per cent of range
- Read up to 4 predefined dynamic variables
- Read or write 8-character tag, 16-character descriptor, date
- Read or write 32 character message
- Read device range, units and damping time constant
- Read or write final assembly number
- Write polling address

Common practice commands

- Read selection of up to 4 dynamic variables
- Write damping time constant
- Write device range
- Calibrate (set zero, set span)
- Set fixed output current
- Perform self-test
- Perform master reset
- Trim PV zero
- Write PV units
- Trim dac zero and gain
- Write transfer function (square root/linear)
- Write sensor serial number
- Read or write dynamic variable assignments

Instrument specific commands

- Read or write low flow cut-off value
- Start, stop or clear totalizer
- Read or write density calibration factor
- Choose PV (mass flow or density)
- Read or write materials or construction information
- Trim sensor calibration

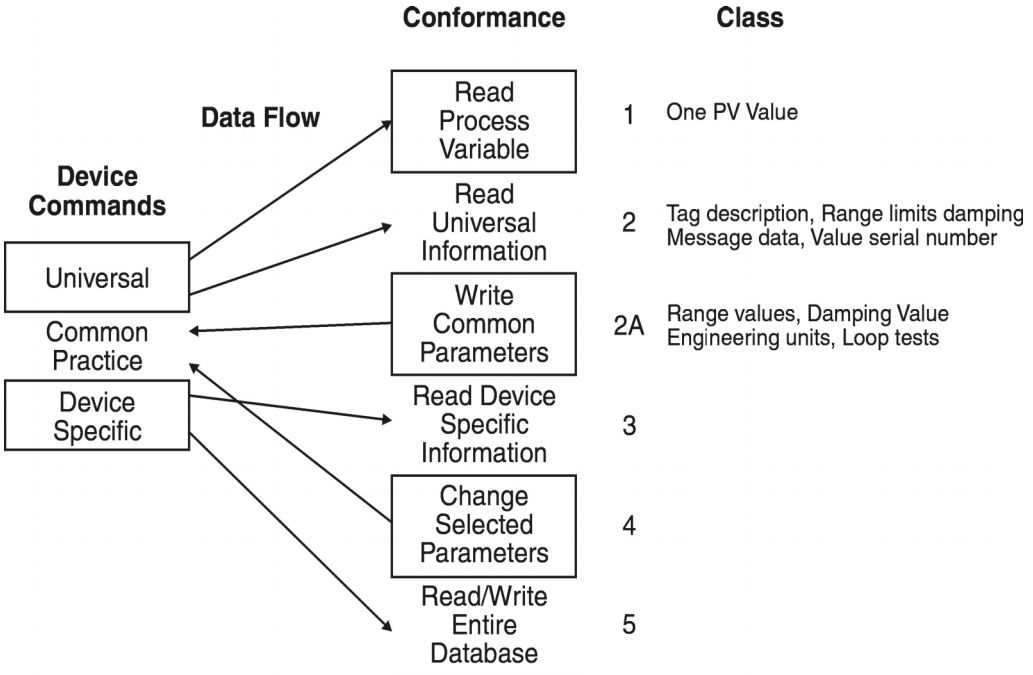


Figure 11.8
HART application layer implementation

Summary of HART benefits

- Simultaneous analog and digital communications
- Allows other analog devices on the highway
- Allows multiple masters to control same smart instrument
- Multiple smart devices on the same highway
- Long distance communications over telephone lines
- Two alternative transmission modes
- Flexible messaging structure for new features
- Up to 256 process variables in any smart field device

11.6 Typical specification for a Rosemount transmitter

Communication specifications

Method of communication:	Frequency shift keying (FSK). Conforms to Bell 202 modem standard with respect to baud rate and binary '1' and binary '0' frequencies.
Baud rate:	1200 bps
Binary '0' frequency:	2200 Hz
Binary '1' frequency:	1200 Hz
Data byte structure:	1 start bit
	8 data bits
	1 odd parity bit
	1 stop bit

Digital process variable rate:	poll/response mode:	2.0 per second
	burst mode:	3.7 per second
No. of multidropped devices:	loop powered:	15 max.
	individually powered:	no limit
Multi-variable specification:	max. 256 process variables per smart device	
Communication masters:	max. 2	

Hardware recommendations

Minimum cable size:	24 AWG, (0.51 mm diameter)
Cable type:	single pair shielded or multiple pair with overall shield
Single twisted pair length:	3048 meters max. (3335 yards)
Multiple twisted pair length:	1524 meters max. (1667 yards)

The following formula can be used to determine the maximum cable length:

$$L = \left[\frac{65\times10^6}{RC}\right] - \left[\frac{Cf + 10000}{C}\right]$$

Where: L = max. length (meters)
R = total resistance (Ω), inclusive of barriers
C = cable capacitance (pF/m)
C_f = max. shunt capacitance of smart field devices (pF)

Worked example

Assume that a Model 3051C smart pressure transmitter, for a Rosemount System 3 control system, is to be installed using a shielded twisted pair. Calculate the maximum cable length permitted for reliable operation.

R = 250 ohms
C = 164 pF/m
C_f = 5000 pF

$$L = \left[\frac{65\times10^6}{250\times164}\right] - \left[\frac{5000+10000}{164}\right]$$

L = 1494 *meters*